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Plant diversity effects on the water balance of an experimental grassland

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ABSTRACT

In the literature, contrasting effects of plant species richness on the soil water balance are reported. Our objective was to assess the effects of plant species and functional richness and functional identity on soil water contents and water fluxes in the experimental grassland of the Jena Experiment. The Jena Experiment comprises 86 plots on which plant species richness (0, 1, 2, 4, 8, 16, and 60) and functional group composition (zero to four functional groups: legumes, grasses, tall herbs, and small herbs) were manipulated in a factorial design. We recorded meteorological data and soil water contents of the 0·0–0·3 and 0·3–0·7 m soil layers and calculated actual evapotranspiration (*ETa*), downward flux (*DF*), and capillary rise with a soil water balance model for the period 2003–2007. Missing water contents were estimated with a Bayesian hierarchical model. Species richness decreased water contents in subsoil during wet soil conditions. Presence of tall herbs increased soil water contents in topsoil during dry conditions and decreased soil water contents in subsoil during wet conditions. Presence of grasses generally decreased water contents in topsoil, particularly during dry phases; increased *ETa* and decreased *DF* from topsoil; and decreased *ETa* from subsoil. Presence of legumes, in contrast, decreased *ETa* and increased *DF* from topsoil and increased *ETa* from subsoil. Species richness probably resulted in complementary water use. Specific functional groups likely affected the water balance via specific root traits (e.g. shallow dense roots of grasses and deep taproots of tall herbs) or specific shading intensity caused by functional group effects on vegetation cover. Copyright © 2013 John Wiley & Sons, Ltd.

KEY WORDS the Jena Experiment; biodiversity; plant diversity; water content; evapotranspiration; soil water flux

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INTRODUCTION

Potential impacts of biodiversity loss on ecosystem functioning have been studied in a number of experiments where biodiversity, often measured as plant species richness, is manipulated as the independent variable (e.g. Tilman *et al*., 1996; Loreau *et al*., 2001; Scherer-Lorenzen *et al*., 2003; Marquard *et al*., 2009). A general result from biodiversity experiments is that increasing plant species richness increases community productivity (Hector *et al*., 1999; Tilman *et al*., 2001; Spehn *et al*., 2005; Marquard *et al*., 2009). As transpiration increases with productivity, this might lead to higher transpiration in diverse mixtures. However, plant cover also increases with species richness (Tilman *et al*., 1996; Spehn *et al*., 2000b), which could compensate the increase in transpiration by stronger shading. The resulting lower soil temperatures with increased shading likely reduce soil evaporation in more diverse mixtures (Spehn *et al*., 2000a).

The complementarity effect including niche differentiation seems to be the most common explanation for the positive plant species richness–productivity relationship in biodiversity experiments (Hector *et al*., 1999; Tilman *et al*., 2001; Spehn *et al*., 2005). Plants can occupy niches with respect to acquisition of many resources including light, nutrients, and water (Tilman *et al*., 1996; Spehn *et al*., 2000b; Verheyen *et al*., 2008). Tilman *et al*. (2012)

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reported that plant productivity in grassland communities was influenced as much by water addition or a major drought as by a change from 4 to 16 plant species. Complementarity among plants with respect to water uptake would also affect the water cycle, which is an important control of nutrient cycles and plant productivity. Soil water contents might decrease because of complementary and more exhaustive water use in species-rich communities. A possible consequence would be reduced downward flux (i.e. loss of water to the deeper soil or groundwater) during wet periods and increased capillary rise (i.e. water input from groundwater) during dry periods.

Effects of plant diversity on evapotranspiration (ET) and soil water contents have been addressed by a number of studies with conflicting results. To our knowledge, there are up to now no published studies where the effect of plant diversity on downward water flux or capillary rise in soil has been investigated. Stocker *et al.* (1999) found no plant species richness effect on ET and also no effect on aboveground biomass, whereas De Boeck *et al.* (2006) and Verheyen *et al.* (2008) reported increasing ET and biomass production with increasing plant diversity. De Boeck *et al.* (2006), Jiang *et al.* (2007), and Mokany *et al.* (2008) reported a negative relationship between plant species richness and soil water contents. Caldeira *et al.* (2001) observed a positive relationship between plant species richness and water contents in 0.0- to 0·15-m soil depths and decreasing soil water contents with increasing plant species richness in 0·15- to 0·3-m depths. Spehn *et al.* (2000a) did not observe any relationship between plant species richness and soil water content in the topsoil (0·06-m soil depth). Rosenkranz *et al.* (2012) and Eisenhauer *et al.* (2013) reported increasing water contents with species richness in the 0·0–0·1 and 0·0–0·06 m soil layers, respectively. In the cited studies, ET and soil water contents were determined at about five measurement dates or from monitoring of approximately 1 month except for De Boeck *et al.* (2006), who collected data from over 1 year in a pot experiment where, however, capillary rise of water from deeper soil layers was prevented.

Jiang *et al.* (2007) and Mokany *et al.* (2008) reported that soil water contents could be explained to a larger degree by functional diversity indices (calculated from plant functional traits included in the plant community) than by species richness. This illustrates that understanding the plant diversity–water balance relationship also requires a thorough consideration of plant functional traits. Biological traits of different plant functional groups such as grasses, legumes, tall herbs, or small herbs (as defined by Roscher *et al.*, 2004) might be the drivers of differences in the water balance of differently diverse plant communities. Water uptake by plants from soil depends on their root distribution and rooting depth (Prasad, 1988; Jarvis, 1989). Grasses might cause a decrease in soil water contents in the topsoil because of their extensive, shallow rooting system and less shading of soil because of their leaf morphology. In a tallgrass prairie, C_3 forbs and shrubs avoided competition under water stress with grasses by using water from deeper soil layers (Nippert and Knapp, 2007). The taproot system of most tall herbs facilitates water use from deeper soil depths, and the large leaves of tall herbs might cause intensive shading. The increase in aboveground community biomass if tall herbs or legumes are present (Marquard *et al.*, 2009) might decrease soil evaporation because of shading but also increase transpiration. Furthermore, the effects of different plant functional groups (e.g. grasses and tall herbs) on water contents and fluxes could compensate each other to a variable extent in plant communities, which would explain conflicting results in previous studies.

Manipulative diversity experiments usually imply a land use change, frequently from arable to grassland. This artificial land use change to a new experimental ecosystem affects different ecosystem properties with likely different timescales (Schilling and Spooner, 2006; Basaran *et al.*, 2008). Shortly after establishment of differently diverse grasslands from seeds, inter-specific and intra-specific competition, complementarity, and niche occupation are the most common factors to explain differences in aboveground biomass production. But there are also slow processes (e.g. development of soil properties in response to changed plant cover) that are often not well understood because of the lack of long-term observations. Steinbeiss *et al.* (2008) reported a decrease in soil C stocks from 2002 to 2004, i.e. during the first 2 years after establishment of a grassland experiment (Jena Experiment) on a former arable field. From 2004 to 2006, there was significant carbon sequestration, which was more pronounced in plots of higher plant species richness (Steinbeiss *et al.*, 2008). The initial decrease was attributed to the land use conversion. Such a more pronounced carbon sequestration in species-rich mixtures was also reported by Fornara and Tilman (2008) for the 6th–12th years of the Cedar Creek biodiversity experiment. Changes in organic matter concentrations are also important for the soil water cycle because of the well-known relationship between organic matter concentrations and the soil water holding capacity. Furthermore, weather conditions might strongly influence the relationship between soil water contents or fluxes and plant diversity, providing an additional explanation for contradicting results in the literature.

The main objective of our study was to investigate the effect of plant diversity on the water balance (soil water content, ET, downward flux, and capillary rise) in two soil layers (0·0–0·3 and 0·3–0·7 m) over 5 years. We hypothesize that (1) plant species richness increases ET and capillary rise and decreases downward flux and water contents because of higher water use of higher biomass,

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(2) the presence of specific plant functional groups affects the water balance because of their specific biological traits (e.g. rooting depth or shading characteristics) and we expect differences between functional group effects on the topsoil (0·0–0·3 m soil layer) and subsoil (0·3–0·7 m soil layer) because of the differing biological traits and the adaptability of water acquisition strategies of plants to avoid competition, and (3) plant diversity effects on the water balance change over time because of climatic differences between different years and because of the proceeding establishment of grassland on former arable land.

METHODS

This study was conducted as part of the Jena Experiment (www.the-jena-experiment.de), which is a grassland plant diversity experiment addressing the role of plant diversity for element cycling and trophic interactions (Roscher *et al*., 2004).

Study site

The field site is located close to the city of Jena, Germany (50°55′N, 11°35′E; 130 m above sea level) on the floodplain of the Saale river. Mean annual air temperature is 9·3 °C, and mean annual precipitation is 587 mm (1961–1990, Kluge and Müller-Westermeier, 2000). The soil is an Eutric Fluvisol that developed from up to 2-m-thick loamy fluvial sediments, almost free of stones. As a result of the fluvial dynamics, the texture ranges from sandy loam near the river to silty clay with increasing distance from the river. This systematic variation in soil texture is considered in the experimental design as the plots are arranged in four blocks that are parallel to the river Saale and have approximately homogeneous soil texture. The site was converted from grassland to an arable field in the 1960s and consequently fertilized and ploughed for crop production until the beginning of the grassland plant diversity experiment in 2002 (Roscher *et al*., 2004).

The entire experimental design is described by Roscher *et al*. (2004). Briefly, the main experiment comprises 86 plots (20 m × 20 m) grouped in four blocks (considering the systematic variation in soil texture). The 86 plots were established from seeds in May 2002 with different levels of species richness (0, 1, 2, 4, 8, 16, or 60 plant species) and functional group richness (zero, one, two, three, or four plant functional groups out of grasses, small herbs, tall herbs, and legumes) chosen by the random replacement method from a species pool of 60 species that are part of Molinio-Arrhenatheretea meadows, Arrhenatherion community (Ellenberg, 1996). Each level of species richness is replicated four times per block, resulting in 16 plots per richness level except for the 0-species and 60-species plots, which were replicated once per block, i.e. four times in the whole experiment. As there was a strong correlation between the number of sown species and the realized species richness ($R^2 > 0{\cdot}9$ in each year during 2003–2007; Marquard *et al*., 2009), the successful establishment of the species richness gradient can be assumed. The management of all plots was adapted to extensive meadows used for hay production and mown twice a year in June and September. The plots were weeded regularly to maintain the sown species composition. During the experimental period, the plots were not fertilized.

Field measurements and data preprocessing

We measured soil water contents [S_P (vol.%)] with frequency domain reflectometry (FDR) using a mobile manual FDR probe (PR1/6, Delta-T-Devices, Cambridge, Great Britain) on all 86 plots in 1–2 weekly resolution between 11 July 2002 and 19 December 2002, 27 March 2003 and 16 September 2004, and 15 April 2005 and 18 January 2006 in 0·1-, 0·2-, 0·3-, 0·4-, and 0·6-m soil depths. The FDR probes were calibrated using measured matrix potentials and soil moisture retention curves of selected plots (Kreutziger, 2006). Water contents were converted from vol.% ($=\mathrm{m}^3\,\mathrm{m}^{-3}\cdot 100$) to millimetre ($=\mathrm{l\,m}^{-2}$) by multiplication with 3 (for the 0·0–0·3 m soil layer) and 4 (for the 0·3–0·7 m soil layer), respectively. At the central automatic meteorological station on the field site, air temperature (*T*), precipitation (*P*), wind velocity (*wv*), relative humidity (*rh*), global radiation (*Rg*), and soil water content [S_C (vol.%)] in 0·08-, 0·16-, 0·32-, and 0·64-m soil depths were available in 10-min resolution between 1 July 2002 and 31 December 2007. In each depth, S_C was measured with one Theta Probe soil moisture sensor – ML2x (Delta-T-Devices, Cambridge, Great Britain).

Soil water contents per plot were aggregated to depth-weighted means for the 0·0–0·3 and 0·3–0·7 m soil layers. Soil water content measurements from the central meteorological station were aggregated to daily depth-weighted means for the 0·0–0·3 and 0·3–0·7 m soil layers. *T* smaller than −30 °C and *rh* smaller than 10% were set to not available (NA). *T*, *wv*, and *rh* were aggregated to daily means and *P* and *Rg* to daily sums. Daily sums of *Rg* equal to 0 were also set to NA. Summary statistics can be found in Table I.

Daily potential ET (*ETp*) was estimated for short grass with the Penman–Wendling equation (DVWK, 1996). Daily means of *T*, *wv*, and *rh*; daily sums of *Rg* and *P*; and the daily astronomically possible sunshine duration were used to calculate *ETp*. To obtain a complete time series of *ETp*, missing values of *T*, *Rg*, and *rh* were linearly interpolated, missing *wv* was substituted with the mean wind velocity of all measured values, and missing *P* was substituted with zero. These substitutions should hardly affect the final results because the number of missing values of *T*, *Rg*, *rh*, *wv*, and *P* was small (Table I).

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Table I. Summary statistics of measured soil water contents (S_P) in the 0·0–0·3 and 0·3–0·7 m soil layers per plot (117 measurement dates) and daily soil water contents (S_C) in the 0·0–0·3 and 0·3–0·7 m soil layers, air temperature (T), precipitation (P), wind velocity (wv), relative humidity (rh), and global radiation (Rg) from the central meteorological station between 1 July 2002 and 31 December 2007 and percentage of roots (r) per plot (only vegetated plots) in the 0·0–0·3 m soil layer in July 2006.

	Mean	Minimum	Maximum	*sd*	*n*	Missing
S_P (0·0–0·3 m) (vol.%)	27·4	3·4	53·2	8·2	10 062	822
S_P (0·3–0·7 m) (vol.%)	33·4	18·5	52·5	4·0	10 062	803
S_C (0·0–0·3 m) (vol.%)	26·9	13·8	49·0	7·0	2 010	212
S_C (0·3–0·7 m) (vol.%)	30·9	21·0	46·0	5·6	2 010	170
T (°C)	9·9	−12·2	26·8	7·5	2 010	10
P (mm)	1·6	0·0	58·0	4·1	2 010	14
wv (m s^{-1})	1·9	0·5	7·1	1·0	2 010	180
rh (%)	78·2	48·9	100·0	9·7	2 010	10
Rg (W m^{-2})	1038	14	3045	776	2 010	13
r (0·0–0·3 m) (%)	80·3	52·0	96·7	9·1	82	2

In July 2006, root biomass was determined by taking three 1·5-m-deep soil cores (87-mm diameter) per plot. Soil cores were separated in depth intervals of 0·0–0·3, 0·3–0·6, 0·6–0·9, 0·9–1·2, and 1·2–1·5 m and cut until roots were less than 10 mm in length. For root extraction, a subsample of soil (0·0–0·3 m: 210 g; 0·3–0·6 m: 600 g) was washed through a sieve (0·5-mm mesh). Collected roots were cleaned, dried (24 h, 70 °C) and weighed. We calculated the percentage of roots in the 0·0–0·3 m soil layer per plot. As this percentage was used to distribute *ETp* to the different soil layers, the percentage of roots in the 0·0–0·3 m soil layer on bare plots was set to 100%, i.e. ET on bare plots can only originate from the 0·0–0·3 m soil layer. We assumed the remaining proportion of roots to extract water from the 0·3–0·7 m soil layer, as on average already 80% of roots were found in the 0·0–0·3 m soil layer (Table I). The two missing root measurements were replaced with the mean percentage of roots in the 0·0–0·3 m soil layer of the vegetated plots.

Modelling soil water contents

As the measured plot-specific soil water content dataset shows partly unequal measurement intervals and some large data gaps, especially in winter 2002/2003 and autumn and winter 2004/2005, we developed a model that is capable of simulating soil water contents in the 0·0–0·3 and 0·3–0·7 m soil layers per plot in (at most) daily resolution. This model should allow the creation of a complete dataset with equidistant timesteps for further analyses and calculations as for instance an annual water budget. For model development, a Bayesian statistical approach was chosen because it allows for missing values in the explanatory variables, considers measurement uncertainty, and needs less parameterization effort than a process-oriented model. We used the program OpenBUGS (Lunn *et al.*, 2009) to set up a Bayesian hierarchical model for each soil depth (0·0–0·3 and 0·3–0·7 m). Linear regression-type models per plot (p) with soil water content measurements (S_C) per timestep (t) from the central field station as explanatory variable [Equation (1)] turned out to be most suitable for the plot-specific water contents (S_P) in this experiment. A normal distribution with mean (μ) 0 and variance (sd^2) 1000 was assumed for parameters a and b [Equations (2) and (3)]. We accounted for the missing soil water content measurements from the central meteorological station (S_C) with a simple submodel that was hierarchically incorporated in the main water content model [Equation (4)]. To ensure that only reasonable values are estimated for the missing S_C values, a uniform distribution with minimum 0 and maximum 100 was assumed for μ_{S_C} [Equation (4)].

$$S_{P_{t,p}} \sim Normal\left(\mu_{S_P} = a_p + b_p \cdot S_{C_t}, sd_{S_P} \sim Uniform(0, 1000)\right) \quad (1)$$

$$a_p \sim Normal\left(\mu_a = 0, sd_a^2 = 1000\right) \quad (2)$$

$$b_p \sim Normal\left(\mu_b = 0, sd_b^2 = 1000\right) \quad (3)$$

$$S_{C_t} \sim Normal\left(\mu_{S_C} \sim Uniform(0, 100), sd_{S_C} \sim Uniform(0, 100)\right) \quad (4)$$

The same model structure was used to simulate water contents (S_P) in the 0·0–0·3 and 0·3–0·7 m soil layers with the centrally measured soil water contents (S_C) of the corresponding depth as explanatory variable.

Convergence of both models occurred at approximately 2000 updates. For each model, convergence was checked using three chains with different random starting values and the modified Gelman–Rubin statistic, as implemented in OpenBUGS. To be conservative, we used 20 000 iterations as 'burn-in'. For each model, one of the chains was run for further 40 000 iterations on which the estimates of the respective water contents and parameters are based. We simulated water contents in a weekly resolution for

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each depth for the period 4 July 2002–27 December 2007 and used this complete dataset with equidistant timesteps for further analyses and calculations.

Model quality was assessed using several quality measures. We calculated the Bayesian coefficient of determination R^2_B, which depicts the proportional reduction of uncertainty obtained through inclusion of the explanatory variables in the model (Ntzoufras, 2009). A Bayesian *p*-value (Gelman *et al.*, 1996) that quantifies the proportion of times when the lack of fit of a replicated data set is greater than the lack of fit of the actual data set was calculated and therefore would have 0·5 as ideal value (Kéry, 2010). Furthermore, we determined average or mean error, mean absolute error, and root mean squared error (Janssen and Heuberger, 1995) from the observed versus mean simulated water contents per depth, calculated a linear regression between the observed and simulated values per depth, and determined the Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) per plot and depth. NSE is a goodness-of-fit measure that ranges from $-\infty$ to 1 with NSE = 1 indicating a perfect fit.

Soil water balance model for water fluxes

We used a soil water balance model, developed by Kreutziger (2006) for our study site, with a few modifications to calculate water fluxes (in mm, i.e. $1\,m^{-2}$) per plot (Figure 1). Because we are aware of the relatively simple assumptions of this water balance model, we compared the results of the water balance model with those of the physically based hydrological model WAVE (Vanclooster *et al.*, 1994) for 21 plots of the Jena Experiment in the earlier work of Kreutziger (2006). The water fluxes calculated with the two different approaches agreed well with each other (Kreutziger, 2006). The soil water balance model has been applied and also described by Oelmann *et al.* (2007a). It is based on

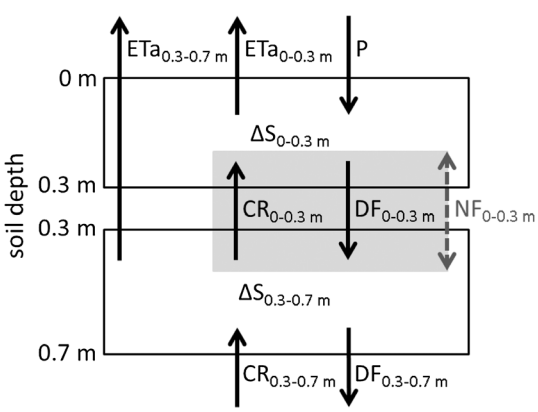

Figure 1. Conceptual diagram of the water fluxes in the studied grassland with actual evapotranspiration (*ETa*), precipitation (*P*), change in soil water content between two sampling dates (ΔS), capillary rise (*CR*), downward flux (*DF*), and net flux ($NF = DF - CR$).

$$P + CR = DF + ETa - \Delta S \tag{5}$$

where P is precipitation, CR is capillary rise, DF is downward flux, ETa is actual ET, and ΔS is the change in volumetric soil water content between two subsequent observation dates ($\Delta S = S_{t_1} - S_{t_2}$). Interception losses were considered negligible in this grassland experiment. We define CR as the capillary rise of water that does not include water transported by roots. The original model had to be modified to allow the calculation of water fluxes for the two soil layers and the interaction between them while considering the influence of the root distribution on ETa from each soil layer. The interaction between the layers was realized by using the net flux ($NF = DF - CR$) of the top layer as input (I) into the bottom layer, i.e. in the 0·0–0·3 m soil layer, I equals P and in the 0·3–0·7 m soil layer, I equals NF. In contrast to P, NF can be negative if CR exceeds DF, which was accounted for in the modifications [Equation (6)]. The percentage of roots (r) in each layer was used as a proxy for the percentage of ETp that could be evaporated from the respective layer [Equations (7) and (8)].

$$\text{if } \Delta S + I < 0 \text{ then } ETa = 0,\ DF = 0,\ CR = -(\Delta S + I) \tag{6}$$

$$\text{if } \Delta S + I \le r \cdot ETp \text{ then } ETa = \Delta S + I,\ DF = 0,\ CR = 0 \tag{7}$$

$$\text{if } \Delta S + I > r \cdot ETp \text{ then } ETa = r \cdot ETp,\ DF = \Delta S + I - r{\cdot}ETp,\ CR = 0 \tag{8}$$

We applied the model to up to the 0·7-m soil depth, but it is possible that if the boundary conditions of Equation (7) apply, additional water, originating from below 0·7 m, undergoes evapotranspiration. We calculated weekly water fluxes ($mm\,week^{-1}$) for each depth and plot using the simulated soil water contents and subsequently calculated annual fluxes ($mm\,year^{-1}$) for 2003–2007.

Statistical analyses

We tested for differences between bare and vegetated plots in mean water contents and mean annual water fluxes per plot using a *t*-test. Subsequently, the bare plots were excluded from further analyses because the water balance of bare plots differs strongly from the water balance of vegetated plots (Section on Effects of Bare Plots and Species Richness on the Soil Water Balance). Monthly mean water contents and annual water fluxes in the 0·0–0·3 and 0·3–0·7 m soil layers of 82 plots (i.e. the four bare plots were excluded) between 2003 and 2007 were analysed for plant diversity effects using repeated-measures analyses of variance (ANOVAs, type I). In each ANOVA, we accounted

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for the block effect before the plant diversity effects, which were included in the following order: species richness and presence/absence of tall herbs, grasses, legumes, and small herbs. Moreover, we calculated additional repeated-measures ANOVAs (type I) for each water content and flux using block, species richness, and functional group richness (instead of presence/absence of individual functional groups) as explanatory variables to test if functional group richness affects water contents and fluxes. To fit the requirements of the ANOVA, soil water contents and *DF* from both depths had to be log-transformed and *CR* from both depths had to be $\log(x+1)$-transformed. We also checked if the differences in variance between the plant diversity treatments were changed through modelling, which could influence the results of the ANOVA. The ratio of simulated variance per treatment to total simulated variance is closely correlated to the ratio of observed variance per treatment to total observed variance (Spearman correlation: $R = 0{\cdot}96$, $p < 0{\cdot}001$), i.e. the model almost did not change the differences in variance between the individual treatments. All statistical analyses were performed with the R 2.15.2 software package (R Core Team, 2012). The *t*-tests were performed with the function *t.test*() and the ANOVAs with *aov*().

RESULTS

Soil water contents and meteorological conditions

The Bayesian *p*-value of the water contents model was close to the ideal value in both soil depths (0·0–0·3 m: *p*-value = 0·51; 0·3–0·7 m: 0·50). Mean error, mean absolute error, and root mean squared error amounted to −0·02, 10·22, and 13·58 mm in the 0·0–0·3 m soil layer and to −0·05, 5·49, and 7·67 mm in the 0·3–0·7 m soil layer, respectively. The model simulated water contents in the 0·0–0·3 m soil layer with $R^2_B = 0{\cdot}68$ and in the 0·3–0·7 m soil layer with $R^2_B = 0{\cdot}76$. A linear regression between observed and simulated water contents in the 0·0–0·3 and 0·3–0·7 m soil layers (Figure 2) resulted in $S_{P_{obs}} = 0{\cdot}06 + 1{\cdot}00 \cdot S_{P_{sim}}$ and $S_{P_{obs}} = 1{\cdot}31 + 0{\cdot}99 \cdot S_{P_{sim}}$, respectively. The plot-specific quality ranged from a NSE of 0·08 to 0·88 (mean NSE = 0·64) for the 0·0–0·3 m soil layer and from −0·00 to 0·83 (mean NSE = 0·60) for the 0·3–0·7 m soil layer. Of the 86 plots, 78 plots in the 0·0–0·3 m soil layer and 68 plots in the 0·3–0·7 m soil layer had an NSE greater than 0·5.

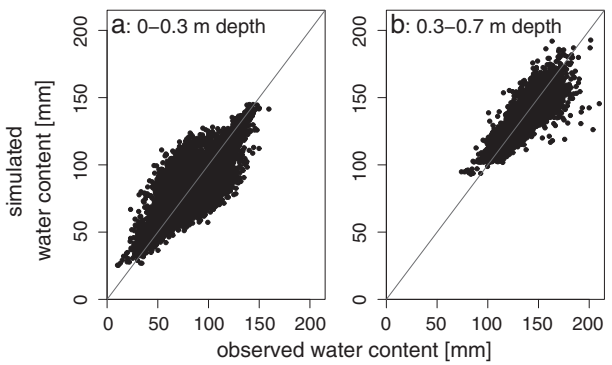

Figure 2. Observed and simulated soil water contents of the 86 plots in the (a) 0·0–0·3 m ($R^2 = 0{\cdot}69$) and (b) 0·3–0·7 m ($R^2 = 0{\cdot}77$) soil layers and the 1–1 line.

Between January 2003 and December 2007, the mean annual precipitation was 533 mm and the *ETp* 658 mm. On average, the annual *ETa* per plot amounted to 374 mm from the 0·0–0·3 m soil layer and 57 mm from the 0·3–0·7 m soil layer, *DF* to 182 and 149 mm, and *CR* to 27 and 48 mm, respectively. Precipitation and mean *DF* were highest in 2007 compared with those in 2003–2006 (Figure 3), and in contrast to 2003–2006, annual precipitation was higher than annual *ETp* in 2007, rendering 2007 the wettest year of the investigated period (Figure 3). Nevertheless, the simulated weekly soil water contents for 2006 and 2007 were within the range of the measured soil water contents in both soil depths. The model quality indicates that the range of measured soil water contents was well met by the model. Therefore, we expect that the simulated water contents also represent the soil moisture conditions in the years 2006 and 2007 well although there were no soil water content measurements available for February 2006–December 2007.

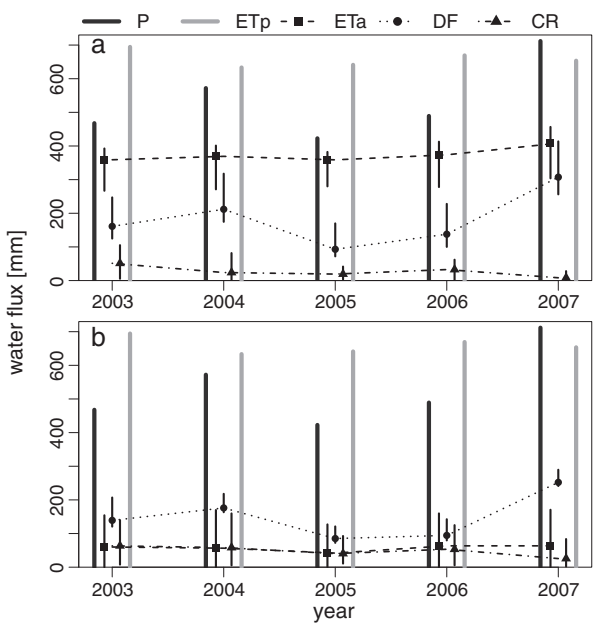

Figure 3. Annual mean fluxes of 86 plots per year and soil layer (a: 0·0–0·3 m; b: 0·3–0·7 m): precipitation (*P*), potential (*ETp*) and actual (*ETa*) evapotranspiration, downward flux (*DF*), and capillary rise (*CR*). *P* and *ETp* are based on the measurements of the central meteorological station and representative for the whole study site. Squares (*ETa*), points (*DF*), and triangles (*CR*) indicate the mean flux over all plots, and ends of lines show minimum and maximum values, respectively. Bars and symbols are shifted with respect to the *x*-axis for clarity.

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Effects of bare plots and species richness on the soil water balance

According to the *t*-test, bare plots had significantly higher mean water contents [0·0–0·3 m: 100 mm ($sd = 2$), $p < 0{\cdot}001$; 0·3–0·7 m: 150 mm ($sd = 8$), $p = 0{\cdot}024$] than vegetated plots [0·0–0·3 m: 93 mm ($sd = 9$); 0·3–0·7 m: 134 mm ($sd = 9$)]. Mean annual *ETa* from the 0·0–0·7 m soil layer was significantly higher ($p < 0{\cdot}001$) on vegetated plots (mean = 432 mm, $sd = 14$) than on bare plots (mean = 399 mm, $sd = 5$). However, *ETa* from topsoil (0·0–0·3 m) alone was higher on bare plots [$p < 0{\cdot}001$; mean vegetated: 373 mm ($sd = 24$), mean bare: 399 mm ($sd = 5$)]. *ETa* cannot originate directly from roots in the 0·3–0·7 m soil layer on bare plots because there are no roots; instead, water needs to be transported by capillary rise (*CR*) from the 0·3-0·7 m soil layer to the 0·0-0·3 m soil layer to be available for *ETa*. *ETa* from the 0·3–0·7 m soil layer of bare plots was hence 0 mm. *DF* from the 0·0–0·3 m soil layer was significantly higher on vegetated plots (mean = 184 mm, $sd = 24$) than on bare plots (mean = 148 mm, $sd = 3$; $p < 0{\cdot}001$), but there was no significant difference in *DF* from the 0·3–0·7 m soil layer. *CR* was significantly higher on vegetated plots [mean in 0·0–0·3 m: 28 mm ($sd = 12$), mean in 0·3–0·7 m: 50 mm ($sd = 23$)] than on bare plots [mean in 0·0–0·3 m: 18 mm ($sd = 3$), mean in 0·3–0·7 m: 20 mm ($sd = 3$)] in both soil layers (0·0–0·3 and 0·3–0·7 m: $p < 0{\cdot}001$). Monthly mean simulated water contents in both soil layers increased significantly with block number (Table II), i.e. with distance from the river and therefore decreasing sand content.

Species richness (Table II) and functional group richness ($p > 0{\cdot}70$) did not significantly change water contents in both soil layers, but the effect of species richness on water contents in the 0·3–0·7 m soil layer changed over time (Table II). During wet soil conditions like in February (on average, the wettest month), species richness significantly decreased water contents in the 0·3–0·7 m soil layer, but in September (on average, the driest month), species richness did not influence water contents (Figure 4). Moreover, mean water contents in the 0·3–0·7 m soil layer under 60 species mixtures varied less between the wettest and driest soil conditions than under mixtures with lower species richness.

Species richness (Table III) and functional group richness (($p > 0{\cdot}42$); analysed in separate repeated-measures ANOVAs as described in the Section on Statistical Analyses) did not have significant effects on the statistically analysed annual water fluxes *ETa*, *DF*, and *CR* in both depths.

Functional group identity effects on the water balance

Water contents were marginally significantly increased in the 0·0–0·3 m soil layer by the presence of tall herbs (Table II). The presence of grasses significantly decreased water contents in the 0·0–0·3 m soil layer (Table II). In the 0·3–0·7 m soil layer, the presence of small herbs increased water contents (Table II). The effect of tall herbs on water contents changed over time in both soil layers, and the effect of grasses on water contents changed over time in the 0·0–0·3 m soil layer (Table II). The difference in mean water contents between plots with and without tall herbs correlated significantly with mean water contents in both soil depths (0·0–0·3 m: $R = -0{\cdot}99$, $p < 0{\cdot}001$; 0·3–0·7 m: $R = 0{\cdot}99$, $p < 0{\cdot}001$). During dry conditions, tall herbs increased water contents in the 0·0–0·3 m soil layer even more than under wet conditions, but the decreasing effect of tall herbs in the 0·3–0·7 m soil layer was less pronounced under dry conditions (Figure 5a, b). The difference in mean water contents between plots with and

Table II. Results of the repeated-measures analysis of variance (type I) for log-transformed monthly mean simulated water contents in the 0·0–0·3 and 0·3–0·7 m soil layers of 82 plots (without bare plots).

	df	*SS*	*F*	*p*
0·0–0·3 m soil layer				
Between-subject effects				
Block	1	5·34	10·51	↑**
Species richness	1	0·82	1·62	0·21
Presence of tall herbs	1	1·92	3·78	↑•
Presence of grasses	1	3·81	7·50	↓**
Presence of legumes	1	0·01	0·01	0·91
Presence of small herbs	1	0·30	0·59	0·45
Residuals	75	38·15		
Within-subject effects				
Month	59	254·54	1615·77	***
Month * presence of tall herbs	59	0·74	4·69	***
Month * presence of grasses	59	0·32	2·03	***
Residuals	4425	11·82		
0·3–0·7 m soil layer				
Between-subject effects				
Block	1	2·58	12·65	↑***
Species richness	1	0·11	0·52	0·47
Presence of tall herbs	1	0·48	2·35	0·13
Presence of grasses	1	0·30	1·48	0·23
Presence of legumes	1	0·02	0·09	0·76
Presence of small herbs	1	2·23	10·94	↑**
Residuals	75	15·27		
Within-subject effects				
Month	59	22·16	673·34	***
Month * species richness	59	0·08	2·27	***
Month * presence of tall herbs	59	0·12	3·60	***
Residuals	4425	2·47		

Nonsignificant within-subject effects are excluded from the table. Arrows indicate an increase (↑) or decrease (↓) of water contents.
• :$p < 0{\cdot}10$, *:$p < 0{\cdot}05$, **:$p < 0{\cdot}01$, ***:$p < 0{\cdot}001$.

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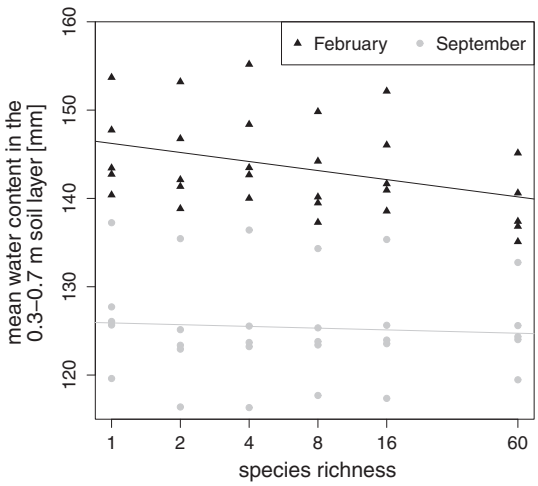

Figure 4. Mean simulated water contents per species richness level in the 0·3–0·7 m soil layer in the, on average, wettest month, February (black triangles), and the, on average, driest month, September (grey dots), of the years 2003–2007. For February, the linear regression between mean water content and log-transformed species richness was significant ($F = 4{\cdot}75$, $p = 0{\cdot}04$) with an R^2 of 0·14 (intercept: 146·23, slope: −1·48), and for September, it was not significant ($F = 0{\cdot}13$, $p = 0{\cdot}73$, $R^2 = 0{\cdot}00$; intercept: 125·90, slope: −0·29).

without grasses correlated significantly with mean water contents in the 0·0–0·3 m soil layer ($R = -0{\cdot}99$, $p < 0{\cdot}001$).

In the 0·0–0·3 m soil layer, *ETa* significantly increased if grasses were present and decreased if legumes were present. Both relationships reversed in the 0·3–0·7 m soil layer (Table III). The presence of small herbs also increased *ETa* in the 0·0–0·3 m soil layer. In both soil layers, the effects of grasses and legumes on *ETa* changed over time and were less pronounced in years with low mean water content (Figure 6a–d). The difference in mean *ETa* between plots with and without grasses (0·0–0·3 m: $R = 0{\cdot}90$, $p < 0{\cdot}05$; 0·3–0·7 m: $R = -0{\cdot}89$, $p < 0{\cdot}05$) as well as with and without legumes (0·0–0·3 m: $R = -0{\cdot}93$, $p < 0{\cdot}05$; 0·3–0·7 m: $R = 0{\cdot}83$, $p = 0{\cdot}08$) correlated with mean annual water content in the considered soil depth. In the 0·3–0·7 m soil layer, the difference in mean *ETa* between plots with and without tall herbs ($R = 0{\cdot}52$, $p = 0{\cdot}37$) as well as with and without small herbs ($R = -0{\cdot}64$, $p = 0{\cdot}24$) did not correlate significantly with mean water contents. The presence of tall herbs and the presence of small herbs decreased *ETa* in the 0·3–0·7 m soil layer in 2003–2006. But in 2007, the wettest year in the observation period (Figures 3 and 6), the presence of tall herbs slightly increased *ETa*, whereas the decreasing effect of small herbs on *ETa* became stronger (Figure 6d). *DF* from the 0·0–0·3 m soil layer significantly decreased if grasses were present and increased if legumes were present (Table III). The latter effects were again less pronounced in years with low mean water content (Figure 6a, e), which is also shown by a significant correlation between the difference in *DF* between plots with and without grasses ($R = -0{\cdot}88$, $p < 0{\cdot}05$) or with and without legumes ($R = 0{\cdot}89$, $p < 0{\cdot}05$) and mean water content. Independent of meteorological conditions, the decreasing effect of small herbs on *DF* from the 0·0–0·3 m soil layer became stronger between 2003 and 2007 (Figure 6e). The difference in *DF* from the 0·0–0·3 m soil layer between plots with and without small herbs was not significantly correlated with mean water content ($R = -0{\cdot}74$, $p = 0{\cdot}16$). In the 0·3–0·7 m soil layer, the presence of tall herbs decreased *DF* significantly, and this effect changed over time, i.e. it was strongest at the beginning of the experiment (Figure 6f), but the difference in *DF* between plots with and without tall herbs did not significantly correlate with mean water content ($R = -0{\cdot}31$, $p = 0{\cdot}61$). The effect of tall herbs on *CR* changed over time in both soil layers (Figure 6g, h), but the difference in *CR* between plots with and without tall herbs did also not significantly correlate with mean water content (0·0–0·3 m: $R = 0{\cdot}46$, $p = 0{\cdot}44$; 0·3–0·7 m: $R = -0{\cdot}45$, $p = 0{\cdot}45$).

DISCUSSION

Effects of species richness

Against our expectations, there were no significant species or functional group richness effects on water contents, *ETa*, *DF*, and *CR* (Tables II and III). Although carbon sequestration in the 0·0–0·3 m soil layer was more pronounced in species-rich plots (Steinbeiss *et al.*, 2008), which potentially increases soil water holding capacity, species richness did not significantly affect water contents in the 0·0–0·3 m soil layer. This complies with the observations of Stocker *et al.* (1999) and Spehn *et al.* (2000a) who did not find significant effects of species or functional group richness on water contents and *ETa*, respectively. This could be explained by two equivalent but opposing effects. As biomass is positively correlated with species richness (e.g. Marquard *et al.*, 2009) and transpiration with biomass, transpiration increases with species richness. By contrast, soil evaporation decreases with species richness because the greater vegetation cover in species-rich communities produces more shade (Spehn *et al.*, 2000a). The findings of Spehn *et al.* (2000a) and Rosenkranz *et al.* (2012) that soil temperatures decreased with increasing species richness support the explanation that there is less soil evaporation in species-rich mixtures. The fact that mean annual *ETa* from the 0·0–0·3 m soil layer was higher on bare plots than on vegetated plots emphasizes the importance of soil evaporation. Therefore, we conclude that the contrasting effects of plant species richness on transpiration and evaporation probably counterbalanced each other in the considered soil layers at our study site in Jena under the meteorological conditions of the observation period. However, this

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Table III. Results of the repeated-measures analyses of variance (type I) for annual fluxes [actual evapotranspiration (*ETa*), log-transformed downward flux (*DF*), and log($x+1$)-transformed capillary rise (*CR*)] between 2003 and 2007 from the 0·0–0·3 and 0·3–0·7 m soil layers of 82 plots (without bare plots).

	ETa				*DF*				*CR*			
	df	*SS*	*F*	*p*	*df*	*SS*	*F*	*p*	*df*	*SS*	*F*	*p*
0·0–0·3 m soil layer												
Between-subject effects												
Block	1	4 833	2·24	0·14	1	0·07	1·12	0·29	1	2·01	1·67	0·20
Species richness	1	549	0·25	0·62	1	0·02	0·31	0·58	1	0·05	0·04	0·84
Presence of tall herbs	1	422	0·20	0·66	1	0·00	0·02	0·89	1	1·48	1·23	0·27
Presence of grasses	1	32 144	14·88	↑***	1	1·14	17·07	↓***	1	0·15	0·12	0·73
Presence of legumes	1	24 031	11·13	↓**	1	0·90	13·56	↑***	1	0·10	0·08	0·77
Presence of small herbs	1	12 714	5·89	↑*	1	0·13	1·95	0·17	1	2·93	2·45	0·12
Residuals	75	161 984			75	4·99			75	90		
Within-subject effects												
Year	4	129 085	791·91	***	4	67·30	9891·97	***	4	147·65	1245·35	***
Year * presence of tall herbs	4	49	0·30	0·88	4	0·01	1·13	0·34	4	0·33	2·77	*
Year * presence of grasses	4	1 417	8·69	***	4	0·06	8·92	***	4	0·06	0·48	0·75
Year * presence of legumes	4	1 102	6·76	***	4	0·04	5·92	***	4	0·09	0·77	0·55
Year * presence of small herbs	4	43	0·26	0·90	4	0·02	2·91	*	4	0·15	1·26	0·29
Residuals	300	12 225			300	0·51			300	8·89		
0·3–0·7 m soil layer												
Between-subject effects												
Block	1	707	0·35	0·56	1	0·00	0·04	0·85	1	2·90	2·70	0·10
Species richness	1	1 287	0·64	0·43	1	0·07	1·60	0·21	1	0·22	0·21	0·65
Presence of tall herbs	1	241	0·12	0·73	1	0·29	6·99	↓**	1	2·78	2·58	0·11
Presence of grasses	1	37 343	18·54	↓***	1	0·00	0·11	0·74	1	0·01	0·00	0·94
Presence of legumes	1	34 235	17·00	↑***	1	0·01	0·24	0·62	1	0·22	0·21	0·65
Presence of small herbs	1	3 142	1·56	0·22	1	0·05	1·12	0·29	1	3·09	2·87	↑•
Residuals	75	151 059			75	3·15			75	80·76		
Within-subject effects												
Year	4	28 893	255·22	***	4	66·33	6203·05	***	4	45·49	943·48	***
Year * block	4	92	0·82	0·52	4	0·01	1·26	0·29	4	0·12	2·50	*
Year * presence of tall herbs	4	293	2·59	*	4	0·05	4·47	**	4	0·10	2·15	•
Year * presence of grasses	4	973	8·59	***	4	0·00	0·36	0·84	4	0·07	1·50	0·20
Year * presence of legumes	4	688	6·08	***	4	0·00	0·28	0·89	4	0·02	0·42	0·80
Year * presence of small herbs	4	278	2·45	*	4	0·01	0·67	0·62	4	0·08	1·56	0·19
Residuals	300	8 491			300	0·80			300	3·62		

Nonsignificant within-subject effects are excluded from the table. Arrows indicate an increase (↑) or decrease (↓) of water contents.
• :$p < 0·10$, *:$p < 0·05$, **:$p < 0·01$, ***:$p < 0·001$.

assumed species richness effect on transpiration could exceed the effect on soil evaporation or vice versa if different depth sections of the soil would be investigated. For example, Rosenkranz *et al*. (2012) reported increasing water contents with species richness in the 0·0–0·1 m soil layer at our study site. Furthermore, Caldeira *et al*. (2001) observed that water contents increased in the 0·0–0·15 m soil layer and decreased in the 0·15–0·3 m soil layer with plant species richness. In our study, only the interaction between species richness and time in the 0·3–0·7 m soil layer was significant and showed that water contents in the 0·3–0·7 m soil layer decreased with increasing species richness when soil moisture was high, e.g. in February, which was on average the wettest month (Figure 4). This complies with the expectation that more biomass in diverse mixtures causes more transpiration. But in drier periods (September was on average the driest month), there was no difference in water contents in the 0·3–0·7 m soil layer between the species richness levels (Figure 4). The smaller temporal variation in water contents of soils under 60 species mixtures indicates a complementary and more exhaustive use of water resources and a higher stability of species-rich communities. The variation in species richness effects with meteorological conditions possibly depicts a

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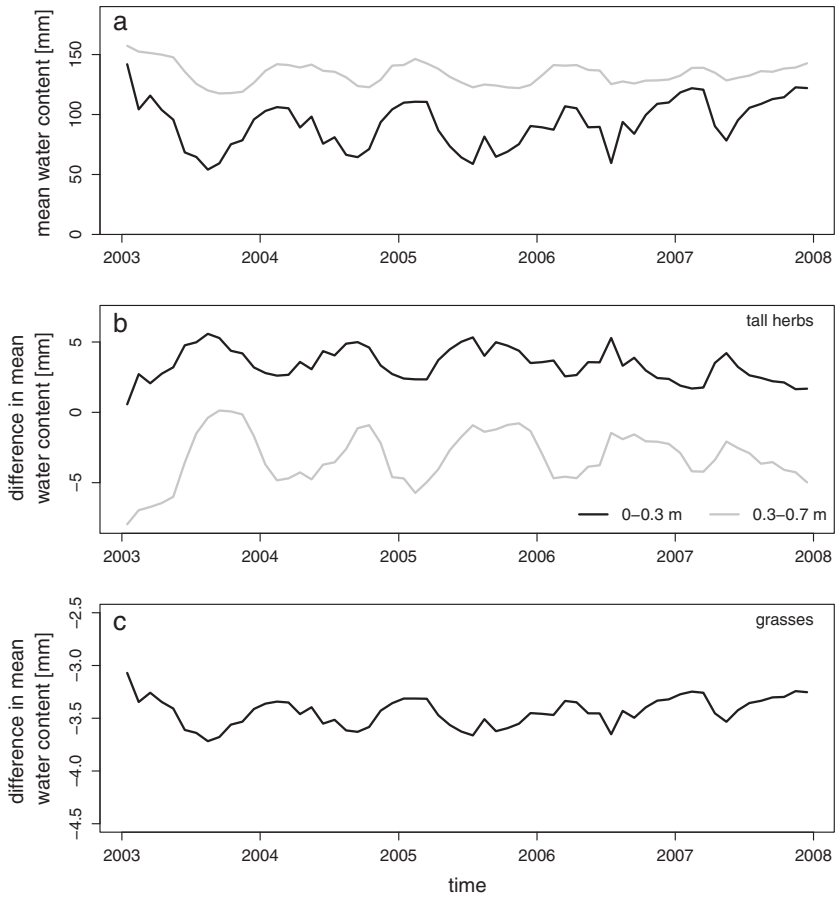

Figure 5. (a) Monthly mean simulated water contents in the 0·0–0·3 m (black line) and 0·3–0·7 m (grey line) soil layers. (b) Difference in mean simulated water content between plots with and without tall herbs in each depth. (c) Difference in mean simulated water content between plots with and without grasses in the 0·0–0·3 m soil layer. Only significant interactions between time and functional group presence/absence are shown.

varying domination of either the shading or increased transpiration effect during specific meteorological conditions. Different meteorological conditions and different depths of the investigated soil layers could explain the conflicting results of previous studies.

Effects of functional group identity

The presence of specific functional groups significantly changed water contents and fluxes with partly opposing effects in the two soil layers. As grasses produce less shade than other plant functional groups because of their leaf morphology, the higher *ETa* and lower water contents in the 0·0–0·3 m soil layer (Tables II and III) may be explained by higher soil evaporation under grass-containing mixtures. Moreover, grasses have a dense, extensive rooting system that facilitates exhaustive water use especially from the upper soil layer and therefore decreases water contents and *DF*. This is in line with the lower *ETa* from the 0·3–0·7 m soil layer of plots with grasses because these plots have proportionately more evapotranspiration from the 0·0–0·3 m and less evapotranspiration from the 0·3–0·7 m soil layer. The increase in *ETa* from the 0·0–0·3 m soil layer if small herbs were present might also be ascribed to less shading because the low canopy height of small herbs suggests a smaller plant cover. The finding that the presence of legumes increased *DF* from the 0·0–0·3 m soil layer and *ETa* from the 0·3–0·7 m soil layer and decreased *ETa* from the 0·0–0·3 m soil layer (Table III) indicates that plant communities including legumes use less water from the 0·0–0·3 m and more from the 0·3–0·7 m soil layer. The higher expected shading effect (Spehn *et al.*, 2000a) because of higher biomass in plots with legumes (Marquard *et al.*, 2009) probably decreased soil evaporation and left more water for *DF* from the 0·0–0·3 m soil layer. In addition, mixtures with legumes had lower root biomass production (Bessler *et al.*, 2012) and standing root biomass in the 0·0–0·3 m soil layer than mixtures without legumes [mean = 539 g m^{-2} (sd = 79) for mixtures with and mean = 782 g m^{-2} (sd = 74) for mixtures without legumes in 2006], which indicates a less intense exploitation of water from this soil layer. The increase in transpiration with biomass might, in contrast, have been satisfied by water uptake from the 0·3–0·7 m soil layer, which would have led to increased *ETa* from this soil layer with the presence of legumes.

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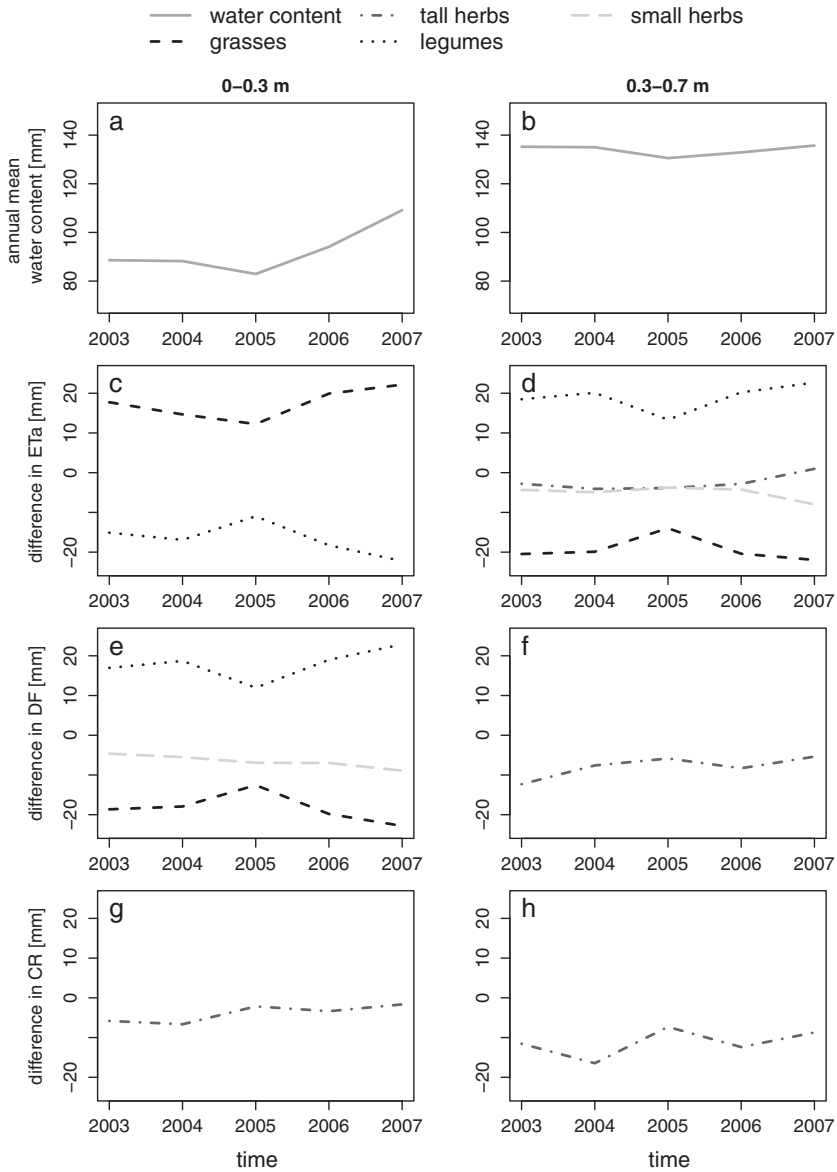

Figure 6. Annual mean simulated water content (grey solid line) in the 0·0–0·3 m (a) and 0·3–0·7 m (b) soil layers. Figures c–h show differences in annual mean water fluxes between plots with and without grasses (black dashed line), tall herbs (grey dash-dotted line), legumes (black dotted line), or small herbs (light grey dashed line), respectively. The shown differences include the difference in annual mean actual evapotranspiration (*ETa*) in the 0·0–0·3 m (c) and 0·3–0·7 m (d) soil layers, the difference in annual mean downward water flux (*DF*) in the 0·0–0·3 m (e) and 0·3–0·7 m (f) soil layers, and the difference in capillary rise (*CR*) in the 0·0–0·3 m (g) and 0·3–0·7 m (h) soil layers. Only significant and marginally significant interactions between time and functional group presence/absence are shown.

Another reason for higher *ETa* and lower water contents in the 0·3–0·7 m soil layer could be that the N_2 fixation ability of legumes increased the amount of plant available N (Oelmann *et al*., 2007b). In a N-limited system, this additional N might have supported plant growth and facilitated the development of deeper roots and the acquisition of resources including water from deeper soil layers for other plant species in the community (Verheyen *et al*., 2008). Tall herbs increased water contents in the 0·0–0·3 m soil layer (Figure 5 and Table II) probably because of a stronger shading effect through increased biomass with the presence of tall herbs (Marquard *et al*., 2009). Another possible explanation could be a higher soil water holding capacity in plots with tall herbs because Steinbeiss *et al*. (2008) reported less carbon loss from the 0·2–0·3 m soil layer of plots with tall herbs between 2002 and 2004. The taproots of most tall herbs, which facilitate water use from deeper soil depths, probably caused the decrease in water contents during wet periods in the 0·3–0·7 m soil layer (Figure 5). The lower *DF* from the 0·3–0·7 m soil layer on plots with tall herbs can therefore be ascribed to the decreased water contents during wet periods in the same

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depth. The lower *DF* together with the decreased *CR* (Figure 6f–h) and the weather-dependent effect on water contents (Figure 5) suggest that tall herbs adapt well to changes in water availability and contribute to a more stable system with a more sustainable water use. This means that tall herbs might use more water from up to 0·7-m depth in times of high water availability and hence decrease *DF*, whereas in times of low water availability, tall herbs use their deep taproots to acquire water from deeper soil layers (i.e. >0·7-m depth) and thus prevent the soil from drying out and thereby decrease *CR*. The negative effect of the presence of tall herbs on *ETa* in the 0·3–0·7 m soil layer in the comparably dry years 2003–2006 and the slightly positive effect on *ETa* in the wettest year 2007 imply less water loss and less dependence on water supply through capillary rise in plant communities that include tall herbs (Figures 3 and 6d).

Temporal variations in plant diversity effects

The result that most plant diversity effects were less pronounced in months or years with low mean soil moisture (Figures 4–6) indicates a dependence of diversity effects on meteorological conditions. The effect of tall herbs on water contents in the 0·0–0·3 m soil layer also depended on meteorological conditions, but it was strongest in months with low soil moisture, which can be assigned to the ability of tall herbs to acquire water from deeper soil depths, especially under water stress. Moreover, the negative effect of tall herbs on *ETa* in the 0·3–0·7 m soil layer in the comparably dry years 2003–2006, which turned slightly positive in the wet year 2007 (Figure 6), points towards a strong dependence of plant diversity effects on meteorological conditions. This change in the effect of tall herbs could not be related to an exceptional performance of tall herbs in biomass production in the wet year 2007 because there was no change in the contribution of tall herbs to total biomass (Marquard *et al.*, 2009). The contrasting results in the literature regarding plant diversity effects on the water balance could therefore be caused by the relatively small datasets that were obtained under specific meteorological and climatic conditions, resulting in either positive, negative, or no diversity effects on water contents or fluxes. Interestingly, the effect of tall herbs on *DF* from the 0·3–0·7 m soil layer and *CR* in both soil layers (Figure 6f–h) became slightly weaker with time since establishment of the experiment, whereas the effect of small herbs on *ETa* from the 0·3–0·7 m soil layer and on *DF* from the 0·0–0·3 m soil layer (Figure 6d, e) became stronger. These effects did not only depend on meteorological conditions and might have therefore been caused by the proceeding establishment of the experimental grassland on former arable land. However, we did not observe a temporal course in plant diversity effects, water contents, or water fluxes that are comparable to previously reported effects of land use conversion from our study site, e.g. decrease in soil C stocks from 2002 to 2004 and carbon sequestration from 2004 to 2006 (Steinbeiss *et al.*, 2008). In addition, we did not find any other diversity effects, besides the aforementioned effects of tall herbs and small herbs, that became stronger or weaker with time since establishment of the experiment, and therefore, we can only speculate that this decrease/increase in effect strength was caused by the proceeding establishment of the grassland ecosystem.

CONCLUSIONS

We conclude the following:

1. The absence of significant species and functional richness effects on the water balance was probably caused by contrasting effects of plant species and functional richness on transpiration and evaporation, which counterbalanced each other in the considered soil layers at our study site under the meteorological conditions of the observation period. However, we found a significant interaction between time and species richness affecting water contents in the 0·3–0·7 m soil layer, which illustrates that meteorological and climatic conditions influence the species richness–water content relationship. Different meteorological and climatic conditions as well as different depths of the investigated soil layers provide possible explanations for contrasting results reported in the literature.
2. The presence of specific plant functional groups influences water contents and fluxes in the 0·0–0·3 and 0·3–0·7 m soil layers according to their specific biological traits. The rooting systems of grasses and tall herbs control the depth and extent to which water is used, and also, the N_2 fixing ability of legumes probably influences the water resource use of the plant community because of better belowground and aboveground growing conditions through improved N availability. Also, differences in biomass and leaf morphology with the presence of specific functional groups and adaptation to environmental conditions probably affect the soil water balance.
3. There was no clear effect caused by the proceeding establishment of grassland on former arable land, but plant diversity effects on the water balance were strongly influenced by meteorological conditions and were usually less pronounced during periods of low soil moisture. In summary, plant diversity effects on water fluxes in grassland cannot be generalized because of the strong influence of meteorological conditions and functional composition of the plant community on the soil water balance.

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